

CSC 360

Operating Systems

Introduction

Wenjun Yang

<https://wenjun-y.github.io/csc360>

Summer 2026



Instructor

- Dr. Wenjun Yang <wenjunyang@uvic.ca>
 - teacher-scholar & postdoc with CS
 - use UVic Teams first, as email not always reliable
 - to help, always include **[csc360]** in your email subject line
 - office hours: **W 12--1pm**
 - or by appointment
 - **on UVic Teams**
 - research area
 - Networked system for distributed AI
 - <https://wenjun-y.github.io/>

CSC360's Mighty TAs



Mostafa Abdollahi



Quanwei Zhang



Ava Birtwistle

Lab/tutorial instructors

- Mostafa, Quanwei, and Ava
 - their email on Brightspace
- Tutorials: start next week!
 - Please attend your registered section only!
 - tutorial lectures
 - C, libc, sockets, pthreads, ...
 - assignment help
 - spec go-through, common problems, ...
 - practice problems

About the course

- *Introduction* to **Operating Systems**
 - (A01/2) **TWF 10:30--11:20am, ECS 125**
 - Bright: “Summer 2026 CSC 360 A01 A02 X”
 - assignments, gradebook, etc
 - **discussion channel hosted on UVic Teams**
 - prerequisites
 - Data structures (CSc **225** or 226)
 - Computer architecture (CSc **230** or CENG 255)
 - System programming (CSc/SENG **265**, CENG)

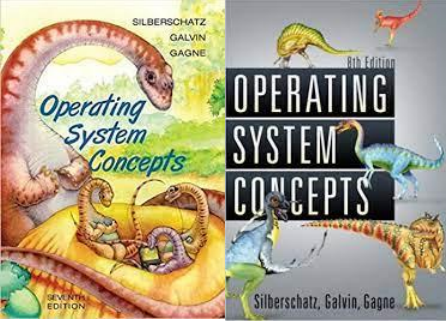
Course evaluation

Assessment for CSC360:		
Assignment 1	5%	Due May 27
Assignment 2	15%	Due June 3
Assignment 3	15%	Due June 30
Assignment 4	15%	Due July 31
Assignments Total	50%	
Midterm 1	10%	June 5
Midterm 2	10%	July 3
Final Exam	30%	TBD
Exams Total	50%	

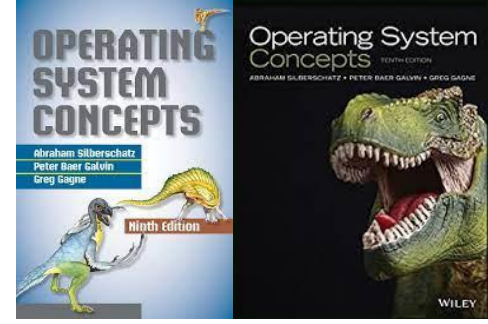
- **Must-pass: A minimum score of 50% on the final exam**

Message from Undergrad Advisor

- Email: cscadvisor@uvic.ca
- Do not have the prerequisite course(s)?
 - need to obtain a waiver
 - otherwise, prerequisite drop after the first week
- Taking the course more than twice?
 - need to have a letter from the Chair and the Dean
 - otherwise, being dropped from the class
- Make sure you can receive email from Bright!
 - use UVic Teams first

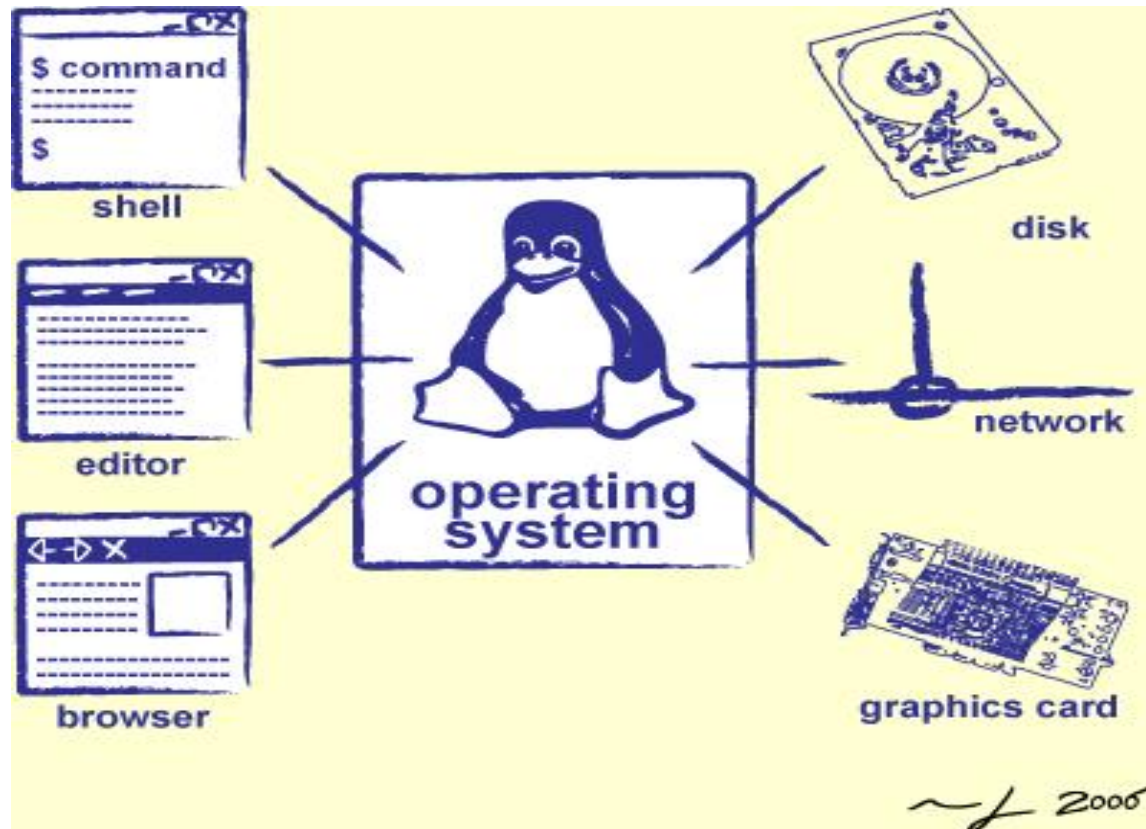


Course materials



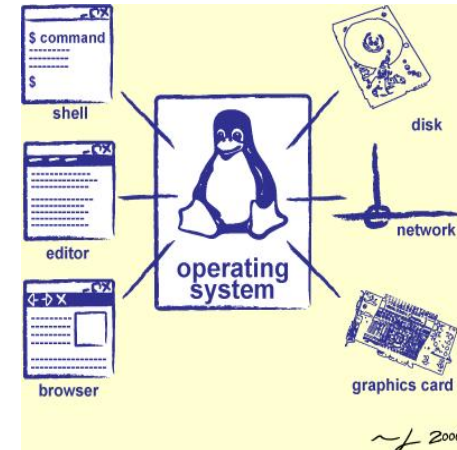
- Required textbook
 - Operating system concepts, 7th or **newer** editions
 - 6th and other editions: different chapter schedules
 - online resources
 - <http://codex.cs.yale.edu/avi/os-book/>
or <http://os-book.com/>
 - errata, slides, practice exercises and solutions
- Explore further
 - Google!

Why take this course?



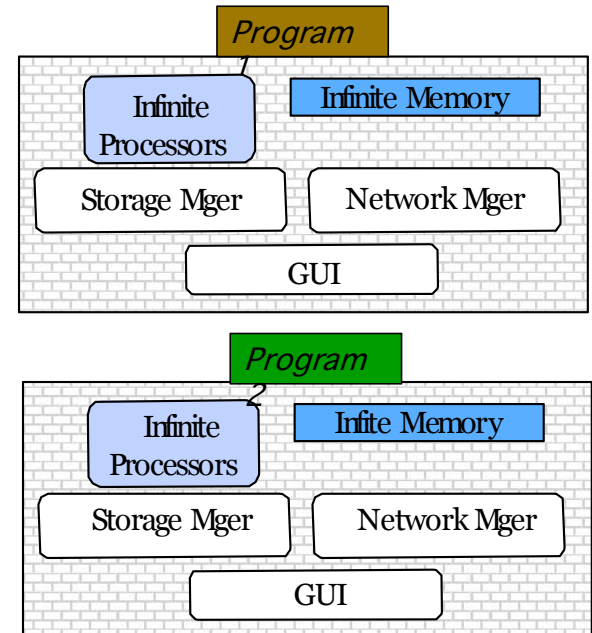
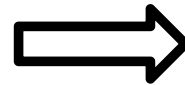
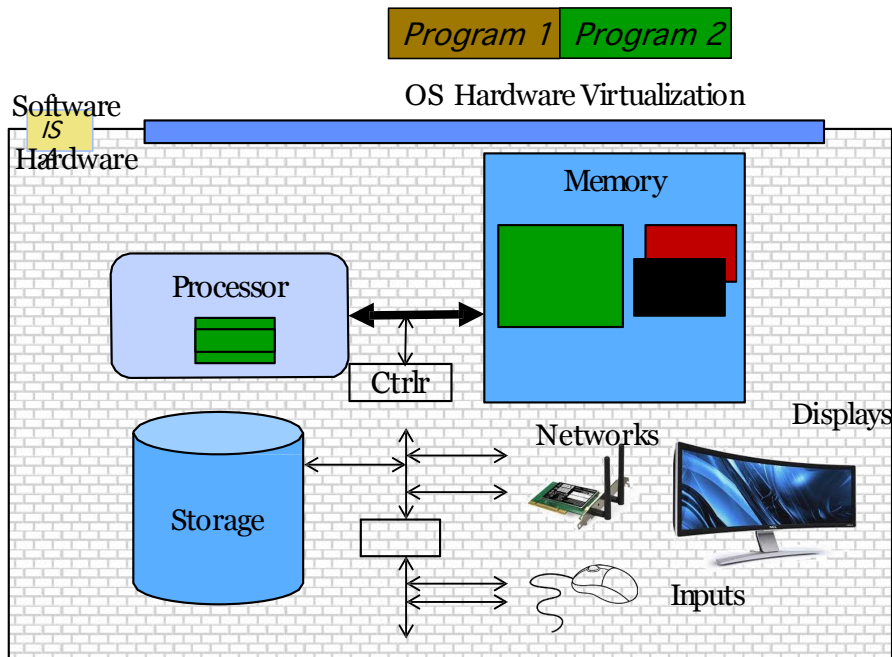
Why take this course?

- The OS is everywhere
 - Every **device**, from your smartwatch, your smart light bulb, to your mobile phone and laptop runs an operating system
 - Every **program** you will ever write will run on an operating system
 - Its **performance** and **execution** behavior will depend on the operating system



What is OS

Referee + illusionist + Glue
=> Easy to use virtual machine



Three aspects for the Class

- Understanding OS principles
- System Programming
- Map Concepts to Real Code

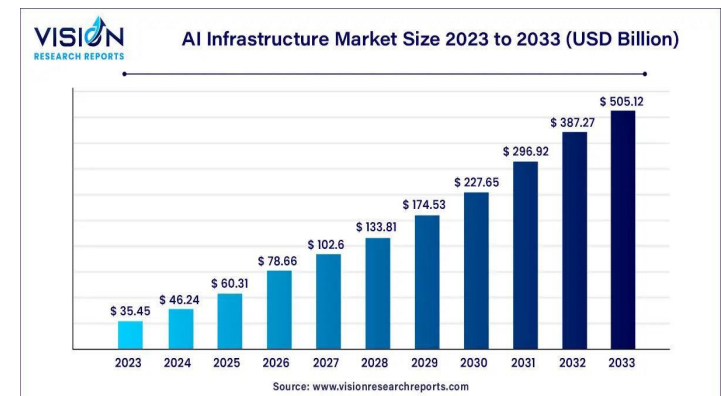
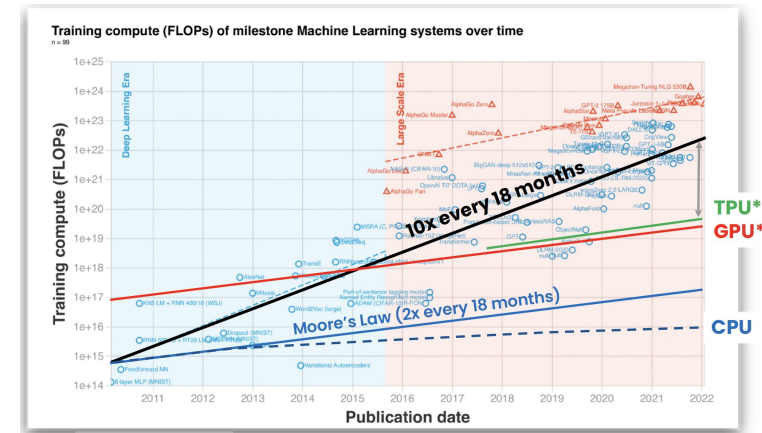
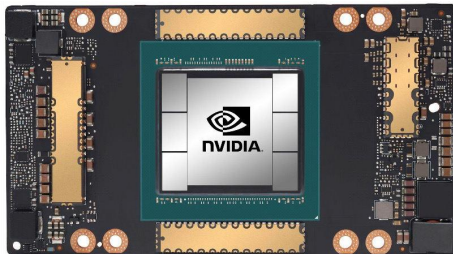
Topic Breakdown (Course Objective)

- virtualizing the CPU (**process**): process, thread, scheduling, synch
- virtualizing memory (**memory**): memory management, virtual memory
- **storage**: file systems, I/O systems

- Distributed system: communication, data processing, etc (not covered but falls into my research focus, happy to chat!)

One last note...

- Operating systems in the age of AI
- Everything gets reinvented
- This class more relevant than ever



Lots of activity over past 15 years!



How do you share a cluster across different distributed workloads ?



How do you share a cluster across different users, jobs, and queries?

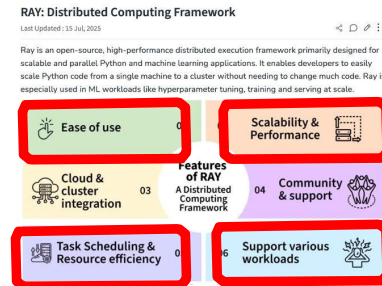
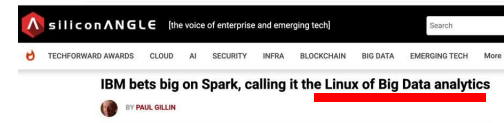


How do you simplify the implementation and execution of distributed AI apps?



How do you do high performance LLM inference for many models across many GPUs supporting different workloads?

Mesosphere unveils Mesos-based datacenter OS plus \$36m injection



- **Memory management**
- **Scheduling, load balancing**
- **Performance isolation**
- **Hierarchical storage (GRAM, RAM, SSD)**
- ...

This lecture so far

- An introduction to the course
 - who, when, where, what, and why
 - course materials
 - course objectives
 - course topics

This lecture so far

- Summary of goals
 - Why should you care?
 - The OS is everywhere
 - Why is it hard?
 - Deal with many different devices, many different time scales.
Safety-critical
 - What is an Operating System?
 - Provides abstraction of a simple, infinite virtual machine
 - Three roles: illusionist, referee and glue
 - A good OS cares about performance, reliability, security and portability

Next lecture

- What is OS
- Interfaces to OS
 - CLI, GUI, system calls, API
 - read OSC7 Chapter 2 (or OSC6 Chapter 3)